

A. Candidate Extraction & vMF Scoring (The Setup)

User Input

Remember to
buy groceries.

a turn may yield one or more candidate
facts; the figure tracks a single ω for clarity

Fact Extraction
Using LLM

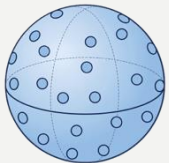
Candidate Fact

c

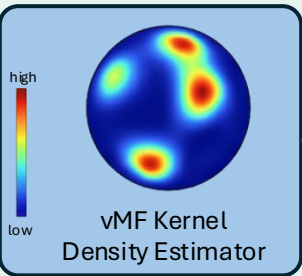
Sentence-Embedding
 l_2 -normalization

Unit vector $\mathbf{c} \in \mathbb{S}^{d-1}$

Unit Hypersphere $\mathbb{S}^{d-1}$



Memory Scope
 $\mathcal{M} = \{\mathbf{m}_1, \dots, \mathbf{m}_N\}$



Novelty Score

$$v(\mathbf{c}) = \frac{1 - s_{\text{VMF}}}{2}$$

B. Adaptive Routing Policy

Adaptive Threshold Computation

PCA projection of
memory vectors
 $\mathbb{S}^{d-1} \rightarrow \mathbb{R}^{d'}$
 $d' \ll d$

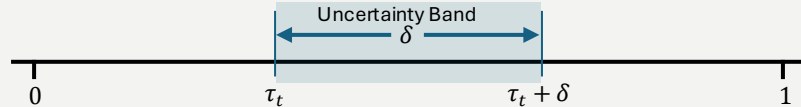
Volume &
Density
(ρ_t)

Density-driven
threshold
 $\tau_t^* = \tau_{\min} + \tau_0 \cdot e^{-\lambda \rho_t}$

EMA Smoothing
 $\tau_t = \alpha \cdot \tau_{t-1} + (1 - \alpha) \tau_t^*$

τ_t

Adaptive Threshold Gate



NOOP ✗
 $v(\mathbf{c}) < \tau_t$
(redundant)

No LLM Call

UPDATE 🧠
 $\tau_t \leq v(\mathbf{c}) < \tau_t + \delta$
(ambiguous)

LLM Call ← the only LLM call

ADD +
 $v(\mathbf{c}) \geq \tau_t + \delta$
(novel)

No LLM Call